

KO SAMUI, Thailand

WMO: 485500

Lat: 9.467N

Lon: 100.050E

Elev: 5

StdP: 101.26

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.1	23.8	19.5	14.3	25.5	20.5	15.1	26.2	8.3	26.4	7.4	27.0	0.5	30	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.4	33.8	26.6	33.2	26.5	32.6	26.5	28.3	31.5	27.9	31.3	27.6	30.9	2.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.3	30.7	27.0	22.7	30.3	26.6	22.2	29.9	91.3	31.7	89.4	31.4	87.7	31.1	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.6	5.0	DB	21.0	35.3	1.9	0.8	19.6	35.9	18.5	36.4	17.4	36.9	16.0	37.4	(4)
(5)			WB	19.4	28.8	1.8	0.9	18.1	29.5	17.0	30.0	16.0	30.5	14.7	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	26.8	27.4	28.1	29.2	29.5	29.2	28.9	28.9	28.5	27.7	27.2	26.9	(6)	
	DBStd	1.39	0.94	0.90	0.98	1.10	1.11	1.15	1.08	1.08	1.02	1.11	0.96	0.92	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6642	522	486	563	575	604	577	585	587	554	550	515	524	(10)	
	CDD18.3	3600	263	253	304	325	346	327	327	329	304	292	265	266	(11)	
	CDH23.3	40952	2661	2766	3634	4181	4344	3967	3809	3824	3401	3055	2683	2626	(12)	
	CDH26.7	14211	564	718	1284	1842	1923	1651	1461	1461	1173	890	658	586	(13)	
(14) Wind	WSAvg	1.5	2.3	2.5	2.1	1.3	1.2	1.1	1.2	1.3	1.3	1.0	1.4	1.8	(14)	
(15) Precipitation	PrecAvg	1980	106	63	109	76	136	126	114	108	116	316	460	250	(15)	
	PrecMax	2872	333	386	836	236	320	255	209	368	232	723	1090	813	(16)	
	PrecMin	1403	16	1	0	2	40	39	39	21	58	63	132	41	(17)	
	PrecStd	399	85	83	181	61	61	57	44	64	46	180	229	173	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.4	31.0	32.1	34.1	34.7	34.5	34.1	34.2	33.7	33.0	31.2	30.7	(19)	
		MCWB	25.7	26.1	26.3	27.0	27.1	27.0	26.4	26.3	26.2	26.3	26.2	25.8	(20)	
	2%	DB	29.6	30.0	31.2	32.5	33.5	33.5	33.1	33.2	32.7	31.7	30.6	30.0	(21)	
		MCWB	25.6	25.8	26.6	27.2	27.0	26.7	26.3	26.2	26.0	26.3	26.2	25.8	(22)	
	5%	DB	29.1	29.5	30.6	31.9	32.7	32.7	32.3	32.5	31.9	30.8	30.1	29.5	(23)	
		MCWB	25.4	25.6	26.5	27.3	27.1	26.5	26.2	26.1	26.0	26.3	26.1	25.6	(24)	
	10%	DB	28.7	29.0	30.1	31.4	32.0	32.0	31.6	31.6	31.1	30.2	29.5	29.1	(25)	
		MCWB	25.3	25.5	26.4	27.1	27.1	26.4	26.1	26.0	26.0	26.2	25.9	25.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	27.0	27.7	28.6	29.1	28.4	27.6	27.7	27.5	27.9	27.3	27.2	(27)	
		MCDB	28.9	29.4	30.3	31.5	31.9	32.4	31.8	31.8	31.0	30.8	29.8	29.2	(28)	
	2%	WB	26.5	26.5	27.2	28.2	28.5	27.7	27.1	27.1	27.0	27.1	26.9	26.7	(29)	
		MCDB	28.6	29.0	30.0	31.3	31.6	31.6	31.2	31.1	30.7	30.1	29.4	28.9	(30)	
	5%	WB	26.1	26.2	27.0	27.8	28.0	27.2	26.8	26.7	26.7	26.7	26.6	26.3	(31)	
		MCDB	28.4	28.7	29.8	31.0	31.3	31.2	30.8	30.7	30.2	29.6	29.0	28.6	(32)	
	10%	WB	25.8	25.9	26.7	27.5	27.6	26.8	26.5	26.4	26.3	26.4	26.3	26.0	(33)	
		MCDB	28.1	28.5	29.5	30.7	31.0	30.8	30.4	30.3	29.8	29.2	28.6	28.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	3.2	3.5	4.3	5.4	5.6	5.6	5.7	5.4	4.8	4.2	4.0	(35)	
		MCDBR	4.2	4.1	4.5	4.9	6.2	6.5	6.6	6.8	6.7	5.8	5.1	4.7	(36)	
	5% WB	MCWBR	2.1	2.0	2.0	2.0	2.3	2.3	2.3	2.3	2.2	2.2	2.3	2.2	(37)	
		MCWBR	3.4	3.3	3.8	4.3	5.2	5.9	5.9	6.1	5.8	5.2	4.4	4.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.489	0.497	0.503	0.469	0.443	0.437	0.438	0.431	0.424	0.449	0.472	0.480	(40)	
		taud	2.207	2.179	2.206	2.314	2.437	2.477	2.459	2.466	2.476	2.402	2.310	2.253	(41)	
	Ebn at Noon	Ebn at Noon	823	830	829	847	853	848	850	868	885	863	834	822	(42)	
		Edh at Noon	145	153	151	134	116	110	112	113	113	121	129	136	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.65	5.53	5.86	5.89	5.32	5.02	4.97	5.10	5.05	4.47	4.01	4.01	(44)	
	RadStd	RadStd	0.43	0.31	0.49	0.38	0.32	0.27	0.34	0.28	0.24	0.41	0.42	0.40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.27	+0.44	N/A	+0.49	+0.28	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.20	+0.35	N/A	+0.36	+0.29	+0.30	N/A	N/A	+74	+73

Nomenclature: See separate page